

Mastery Set: Log → Polynomials — Solutions

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Part 1 — Logarithm Properties

1. $\log_3(81)$

$3^4 = 81 \Rightarrow$ 4

2. $\log_2(1/16)$

$2^{-4} = 1/16 \Rightarrow$ -4

3. $\log_5(125)$

$5^3 = 125 \Rightarrow$ 3

4. $\ln(e^7)$

$\ln(e^x) = x \Rightarrow$ 7

5. $\log(10,000)$

$10^4 = 10,000 \Rightarrow$ 4

6. $\log_4(8) + \log_4(2)$

$= \log_4(16) = 2.$ 2

7. $\log_3(54) - \log_3(2)$

$= \log_3(27) = 3.$ 3

8. Expand $\log_2(x^2 y^3)$

Product rule then power rule: $\log_2(x^2) + \log_2(y^3) = 2 \log_2(x) + 3 \log_2(y).$

$2 \log_2(x) + 3 \log_2(y)$

9. Expand $\ln(\sqrt{a/b})$

Power: $\ln(a/b)^{1/2} = \frac{1}{2} \ln(a/b).$

Quotient: $\frac{1}{2} (\ln a - \ln b).$

$\frac{1}{2} \ln(a) - \frac{1}{2} \ln(b)$

10. Expand $\log_5\left(\frac{x^3y}{z^2}\right)$

Quotient: $\log_5(x^3y) - \log_5(z^2)$. Then product + power: $3\log_5(x) + \log_5(y) - 2\log_5(z)$.

$$3\log_5(x) + \log_5(y) - 2\log_5(z)$$

11. Condense $4\log(x) + 2\log(y) - 3\log(z)$

Reverse power: $\log(x^4) + \log(y^2) - \log(z^3) = \log\left(\frac{x^4y^2}{z^3}\right)$.

$$\log\left(\frac{x^4y^2}{z^3}\right)$$

12. Condense $\frac{1}{2}\ln(a) - 3\ln(b)$

$\ln(a^{1/2}) - \ln(b^3) = \ln\left(\frac{\sqrt{a}}{b^3}\right)$.

$$\ln\left(\frac{\sqrt{a}}{b^3}\right)$$

Part 2 — Log Equations

13. Solve $\log_2(x) = 5$

Domain: $x > 0$.

Convert: $x = 2^5 = 32$. $32 > 0$ ✓.

$$x = 32$$

14. Solve $\log_3(x - 2) = 4$

Domain: $x > 2$.

$x - 2 = 3^4 = 81 \Rightarrow x = 83$. ✓

$$x = 83$$

15. Solve $\log_2(x) + \log_2(x + 2) = 3$

Domain: $x > 0$.

$\log_2(x(x + 2)) = 3 \Rightarrow x(x + 2) = 8$.

$x^2 + 2x - 8 = 0 \Rightarrow (x + 4)(x - 2) = 0$.

$x = 2$ or $x = -4$. Reject -4 (outside domain).

$$x = 2$$

16. Solve $\log_5(x + 1) - \log_5(x - 3) = 1$

Domain: $x + 1 > 0$ AND $x - 3 > 0 \Rightarrow x > 3$.

$\log_5\left(\frac{x + 1}{x - 3}\right) = 1 \Rightarrow \frac{x + 1}{x - 3} = 5$.

$x + 1 = 5(x - 3) = 5x - 15 \Rightarrow -4x = -16 \Rightarrow x = 4$. $4 > 3$ ✓.

$$x = 4$$

Part 3 — Exponential Modeling

17. Solve $5^x = 14$

$x = \log_5(14) = \frac{\ln 14}{\ln 5} = \frac{2.6391}{1.6094} \approx 1.6397$.

$$x \approx 1.6397$$

18. \$3,000 at 5.5% continuous $\rightarrow$ \$9,000?

$$9000 = 3000 e^{0.055 t} \Rightarrow 3 = e^{0.055 t}.$$

$$\ln 3 = 0.055 t \Rightarrow t = \frac{\ln 3}{0.055} = \frac{1.0986}{0.055} \approx 19.975.$$

$$t \approx 19.97 \text{ years}$$

19. $f(t) = 800(1.045)^t \rightarrow 800 e^{rt}$

$$(a) (1.045)^t = e^{t \ln(1.045)} \Rightarrow r = \ln(1.045).$$

(b) Decimal: $r \approx 0.04402$, so continuous rate $\approx 4.40\%$ (slightly less than the discrete 4.5%).

$$(a) r = \ln(1.045) \quad (b) \approx 4.40\%$$

20. 100 g $\rightarrow$ 10 g over 50 yr

$$(a) 10 = 100 e^{50k} \Rightarrow 0.1 = e^{50k} \Rightarrow k = \frac{\ln(0.1)}{50} = -\frac{\ln 10}{50} \approx -0.04605.$$

$$(b) \text{Half-life: } \frac{1}{2} = e^{k t_h} \Rightarrow t_h = \frac{\ln(1/2)}{k} = \frac{-\ln 2}{-\ln 10/50} = \frac{50 \ln 2}{\ln 10} \approx \frac{50(0.6931)}{2.3026} \approx 15.05.$$

$$(a) k = -\frac{\ln 10}{50} \quad (b) t_h \approx 15.05 \text{ years}$$

Part 4 — Fundamental Theorem of Algebra

21. $(3 - i)$ zero of $P(x) = x^4 - 9x^3 + 30x^2 - 42x + 20$

Step 1. Conjugate $(3 + i)$ is also a zero.

Step 2. Quadratic factor: $(x - (3 - i))(x - (3 + i)) = (x - 3)^2 - i^2 = x^2 - 6x + 10$.

Step 3. Divide $P(x)$ by $x^2 - 6x + 10$ (long division):

$$\frac{x^4 - 9x^3 + 30x^2 - 42x + 20}{x^2 - 6x + 10} = x^2 - 3x + 2.$$

Step 4. Factor: $x^2 - 3x + 2 = (x - 1)(x - 2)$.

All zeros: $\{1, 2, 3 - i, 3 + i\}$.

$$P(x) = (x - 1)(x - 2)(x - (3 - i))(x - (3 + i))$$

22. Degree 4, real int coeffs, zeros $x = 2$ (mult 2) & $x = 2 + i$

Step 1. Conjugate $2 - i$ is also a zero. Four zeros: $2, 2, 2 + i, 2 - i$.

Step 2. Complex pair: $(x - (2 + i))(x - (2 - i)) = (x - 2)^2 - i^2 = x^2 - 4x + 5$.

Step 3. $(x - 2)^2 = x^2 - 4x + 4$. Multiply with $x^2 - 4x + 5$:

$$(x^2 - 4x + 4)(x^2 - 4x + 5).$$

Let $u = x^2 - 4x$. Then $(u + 4)(u + 5) = u^2 + 9u + 20$. So:

$$(x^2 - 4x)^2 + 9(x^2 - 4x) + 20 = x^4 - 8x^3 + 16x^2 + 9x^2 - 36x + 20 = x^4 - 8x^3 + 25x^2 - 36x + 20.$$

$$P(x) = x^4 - 8x^3 + 25x^2 - 36x + 20$$

23. Degree 3, real coeffs, zeros $x = -1$ and $x = 3i$

Conjugate $-3i$ also a zero. $(x - 3i)(x + 3i) = x^2 + 9$.

$$P(x) = (x + 1)(x^2 + 9) = x^3 + 9x + x^2 + 9 = x^3 + x^2 + 9x + 9.$$

$$P(x) = x^3 + x^2 + 9x + 9$$

Part 5 — Synthetic Division & Factor Theorem

24. $(2x^3 + 3x^2 - 8x + 3) \div (x - 1)$

$c = 1$. Synthetic:

1		2	3	-8	3
			2	5	-3
		2	5	-3	0

Quotient $Q(x) = 2x^2 + 5x - 3$; remainder $R = 0 \Rightarrow (x - 1)$ IS a factor.

$$Q(x) = 2x^2 + 5x - 3, R = 0. (x - 1) \text{ is a factor.}$$

25. $P(-2)$ for $P(x) = x^4 - 3x^2 + 2x - 5$

Insert zero for x^3 . Synthetic with $c = -2$:

-2		1	0	-3	2	-5
			-2	4	-2	0
		1	-2	1	0	-5

By Remainder Theorem, $P(-2) = -5$. Check directly: $16 - 0 - 12 - 4 - 5 = -5$. ✓

$P(-2) = -5$

26. Find k so $(x - 3)$ is factor of $P(x) = x^3 - 2x^2 + kx - 6$

Factor Theorem: $P(3) = 0$.

$$P(3) = 27 - 18 + 3k - 6 = 3 + 3k = 0 \Rightarrow k = -1.$$

$k = -1$

27. Factor completely: $P(x) = x^3 - 7x + 6$ ($x = 1$ is a zero)

Verify: $P(1) = 1 - 7 + 6 = 0$. ✓

Synthetic with $c = 1$ (insert 0 for x^2):

1		1	0	-7	6
			1	1	-6
		1	1	-6	0

Quotient: $x^2 + x - 6 = (x + 3)(x - 2)$.

$P(x) = (x - 1)(x - 2)(x + 3)$

Part 6 — Polynomial Graphing

28. $P(x) = (x - 1)(x + 2)^2(x - 3)$

Degree: $1 + 2 + 1 = 4$. Leading term $+x^4 \Rightarrow$ end behavior: $\Uparrow$ (both ends to $+\infty$).

Zeros: $x = 1$ (mult 1, cross); $x = -2$ (mult 2, bounce); $x = 3$ (mult 1, cross).

y-intercept: $P(0) = (-1)(2)^2(-3) = (-1)(4)(-3) = 12$. Point $(0, 12)$.

Deg 4, $\Uparrow$; zeros: 1 (cross), -2 (bounce), 3 (cross); y-int 12

29. $P(x) = -x^3(x - 2)(x + 4)^2$

Degree: $3 + 1 + 2 = 6$. Leading term: $-x^3 \cdot x \cdot x^2 = -x^6 \Rightarrow$ end behavior: $\Downarrow$ (both to $-\infty$).

Zeros: $x = 0$ (mult 3, slide / flat-cross); $x = 2$ (mult 1, cross); $x = -4$ (mult 2, bounce).

y-intercept: $P(0) = 0$ (graph passes through origin via slide).

Deg 6, $\Downarrow$; zeros: 0 (slide), 2 (cross), -4 (bounce); y-int 0

30. Build degree-4 with zeros -2 (cross), 0 (bounce), 3 (cross), neg leading coef

Multiplicities: -2 cross $\Rightarrow$ mult 1; 0 bounce $\Rightarrow$ mult 2; 3 cross $\Rightarrow$ mult 1. Total $1 + 2 + 1 = 4$. ✓

Equation (one valid choice): $P(x) = -x^2(x + 2)(x - 3)$.

End behavior: degree 4 even, leading $-1 \Rightarrow \Downarrow$.

Sketch: Coming from $-\infty$ on left, crosses up at $x = -2$, peaks above axis, touches and bounces down off zero at $x = 0$, goes back up briefly, crosses down at $x = 3$, drops to $-\infty$ on right.

(Other valid forms: any negative scalar multiple, e.g. $-2x^2(x + 2)(x - 3)$ etc.)

$P(x) = -x^2(x + 2)(x - 3)$, end $\Downarrow$